CH 6 — Chain Rule

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6.1 Chain Rule

Recall we use the chain rule to take derivatives of composite functions.

Consider

$$T(t) = f(x(t))$$

where $x(t)$ represents the position of an ant crowling along a log and $T(t)$ represents the remperature of the log as the ant moves along.

By the chain rule the change in the temperature can be obtained

$$T'(t) = f'(x(t)) \cdot x'(t)$$

In 2D, if we allow the ant to move in any direction, that is $f(x, y)$ is the temperature at the point (x, y) and $\vec{x}(t) = (x(t), y(t))$ represents the position of the ant as a function of time.

Then,

$$T(t) = f(x(t), y(t))$$

represents the temperature experienced by the ant as it moves along its path.

Let t change by a small amount Δt, then $\begin{cases} \Delta x = x(t+\Delta t) - x(t) \\ \Delta y = y(t+\Delta t) - y(t) \end{cases}$.

By the increment form of linear approximation,

$$\Delta T \approx \frac{\partial T}{\partial x}\Delta x + \frac{\partial T}{\partial y}\Delta y$$

$$\lim_{\Delta t \to 0} \frac{\Delta T}{\Delta t} \approx \lim_{\Delta t \to 0} \left[\frac{\partial T}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial T}{\partial y}\frac{\Delta y}{\Delta t} \right]$$

$$\frac{\mathrm{d}T}{\mathrm{d}t} \approx \frac{\partial T}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial T}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

If $T(x, y)$ is differentiable, we can show that the error in the linear approximation gives 0

> 📘 **Definition — Chain Rule**
>
> Let $T(t) = f(x(t), y(t))$ and $a = x(t_0), b = y(t_0)$. If f is differentiable at (a, b) and both $x'(t_0)$ and $y'(t_0)$ exist. Then $T'(t)$ exists and given by
>
> $$T'(t_0) = f_x(a, b) \cdot x'(t_0) + f_y(a, b) \cdot y'(t_0)$$

Example:

Given $T(x, y) = x^2 + 2y^2, x(t) = e^t \sin t, y(t) = \cos t$, find $T'(t)$

$T'(t) = \left[e^{2t}\sin^2 t + 2\cos^2 t\right]' = 2te^{2t}\sin^2 t + 2e^{2t}\sin t\cos t - 4\cos t \sin t$

Alternatively,

$T'(t) = 2(e^t \sin t)(e^t \sin t + e^t \cos t) + 4(\cos t)(-\sin t) = 2e^{2t} \sin^2 t(\sin t + \cos t) - 4 \cos t \sin t$

6.2 Generalization to $\mathbb{R}^n$

> 📘 **Definition — Generalization of Chain Rule**
>
> The definition of a differentiable fucntion and all the theorems on differentiablility generalize to $\mathbb{R}^n$
>
> Let $\vec{x}(t) \in \mathbb{R}^n, T(t) = f(\vec{x}(t))$, we have
>
> $$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mathrm{d}f(\vec{x}(t))}{\mathrm{d}t} = \nabla f(\vec{x}(t)) \cdot \frac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t}$$

Example

Let $f(x, y, z)$ be a differentiable function and $g(t) = f(x(t), y(t), z(t))$

where $x(t) = t, y(t) = t^2, z(t) = t^3$. If $\nabla f(1, 1, 1) = \left(2, \frac{1}{2}, 1\right)$, find $g'(1)$

$$g'(t) = \nabla f(\vec{x}(t)) \cdot \frac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t} \overset{t=1}{\Longrightarrow} g'(1) = \nabla f(\vec{x}(1)) \cdot \frac{\mathrm{d}\vec{x}(1)}{\mathrm{d}t} = \nabla f(1, 1, 1) \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ \frac{1}{2} \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 6$$

Consider $T(x) = f(x(t), y(t))$

T		Dependent variable
x	y	Intermediate variables
t	t	Independent variables

To find $\frac{\mathrm{d}T}{\mathrm{d}t}$, find all paths which connects T to t. For each path, write down the corresponding derivatives and sum the results.

$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\partial T}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial T}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$

Examples:

1. Let f be a differentiable function s.t $\nabla f(1, 0) = (3, -2)$. and let $z = g(x, y) = f(xy, x^2 - y^2)$. Find $\frac{\mathrm{d}z}{\mathrm{d}x}$ at $(1, 1)$

   Let $u = xy$ and $v = x^2 - y^2$ so that $g(x, y) = f(u, v)$

z			
u		v	
x	y	x	y

   $\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x} = \frac{\partial f}{\partial u}(y) + \frac{\partial f}{\partial v}(2x)$

   Let $(x, y) = (1, 1)$

   $$\left.\frac{\partial z}{\partial x}\right|_{(1,1)} = \frac{\partial f}{\partial u}(u(1,1), v(1,1)) + \frac{\partial f}{\partial v}(u(1,1), v(1,1)) \cdot (2)(1) = \frac{\partial f}{\partial u}(1, 0) + 2\frac{\partial f}{\partial v}(1, 0)$$
   $$= (1)(3) + (2)(-2) = -1$$

2. $z = x^2y^4 - 2y, y = \sin x^2$. Find $\frac{\mathrm{d}z}{\mathrm{d}x}$

Let $u = x^2y^4 \implies z = x^2(\sin x^2)^4 - 2(\sin x^2)$

z	
x	y
	x

$\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = 2y^4 + (4y^3x^2 - 2)2x\cos x^2 = 2\sin^4(x^2) + 8x^3\sin^3(x^2)\cos x^2 - 4x\cos x^2$

3. $u = f(v, w, t)$, where $v = g(x, y, z), w = h(x, y), x = x(t), y = y(t)$. Find $\frac{\partial u}{\partial t}$.

u					
v			w		t
x	y	z	x	y	
t	t		t	t	

$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial v}\left(\frac{\partial v}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial v}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}\right) + \frac{\partial u}{\partial w}\left(\frac{\partial w}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial w}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}\right) + \left(\frac{\partial u}{\partial t}\right)_{v,w}$

Notice the above there are two $\frac{\partial u}{\partial t}$, but they are actually different.

$\frac{\partial u}{\partial t}$ on LHS is the partial derivative of u with respect to t taking into account of all dependence

$\frac{\partial u}{\partial t}$ on the RHS which is can be denoted for holding v, w constant: $\left(\frac{\partial u}{\partial t}\right)_{v,w}$

Chain Rule for Second Order Partial derivatives

We can also use the chain rule to calculate higher order partial derivatives, but it can get a bit complicated.

Examples:

1. $z = f(x(t))$, $\frac{\mathrm{d}^2z}{\mathrm{d}t^2}$

$\frac{\mathrm{d}^2z}{\mathrm{d}t^2} = \frac{\mathrm{d}z}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = f'(x(t))\cdot x'(t) = \frac{\mathrm{d}}{\mathrm{d}x}f'(x(t))\cdot x'(t)$

$= \frac{\mathrm{d}}{\mathrm{d}t}f'(x(t))x'(t) + f'(x(t))x''(t) = f''(x(t))(x'(t))^2 + f'(x(t))x''(t)$

2. $z = g(u, v) = f(x(u, v), y(u, v))$ Find z_{uv}

$z_u = f'(x(u,v), y(u,v))(x'(u,v) + y'(u,v))(u') = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u} = f_x x_u + f_y y_u$

$z_{uv} = \frac{\partial z_u}{\partial v} = \frac{\partial f_x x_u + f_y y_u}{\partial v} = \frac{\partial f_x}{\partial v}x_u + f_x\frac{\partial x_u}{\partial v} + \frac{\partial f_y}{\partial v}y_u + f_y\frac{\partial y_u}{\partial v}$

$= \big(f_{xx}\cdot x_v + f_{yx}\cdot y_v\big)f_x\cdot x_{uv} + \big(f_{yx}\cdot x_v + f_{yy}\cdot y_v\big)f_y\cdot y_{uv}$